

8(a)(i)	at least one anticlockwise arrow and no clockwise arrows	B1
8(a)(ii)	(force is to the) left	B1
8(a)(iii)	force is the same	B1
	Newton's third law (of motion) or force depends on the product of the two currents	B1

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Question	Answer	Marks
8(b)(i)	frequency of radio waves is equal to natural frequency of protons	B1
	resonance of protons occurs / protons absorb energy	B1
8(b)(ii)	in between pulses / when pulse stops	B1
	Any 1 from: • protons de-excite • protons emit r.f. pulses • emitted (r.f.) pulse (from proton) detected	B1

Question	Answer	Marks
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